AI-Assisted Python Projects

Lecture 7 Report — **Syeda Hafsa Fatima Naqvi**

Four practical projects exploring how AI can responsibly solve real problems, with all results manually verified.

Four Projects, One Approach

💰 Money Detective

Categorize expenses and find the highest spending category

📊 What's My Grade?

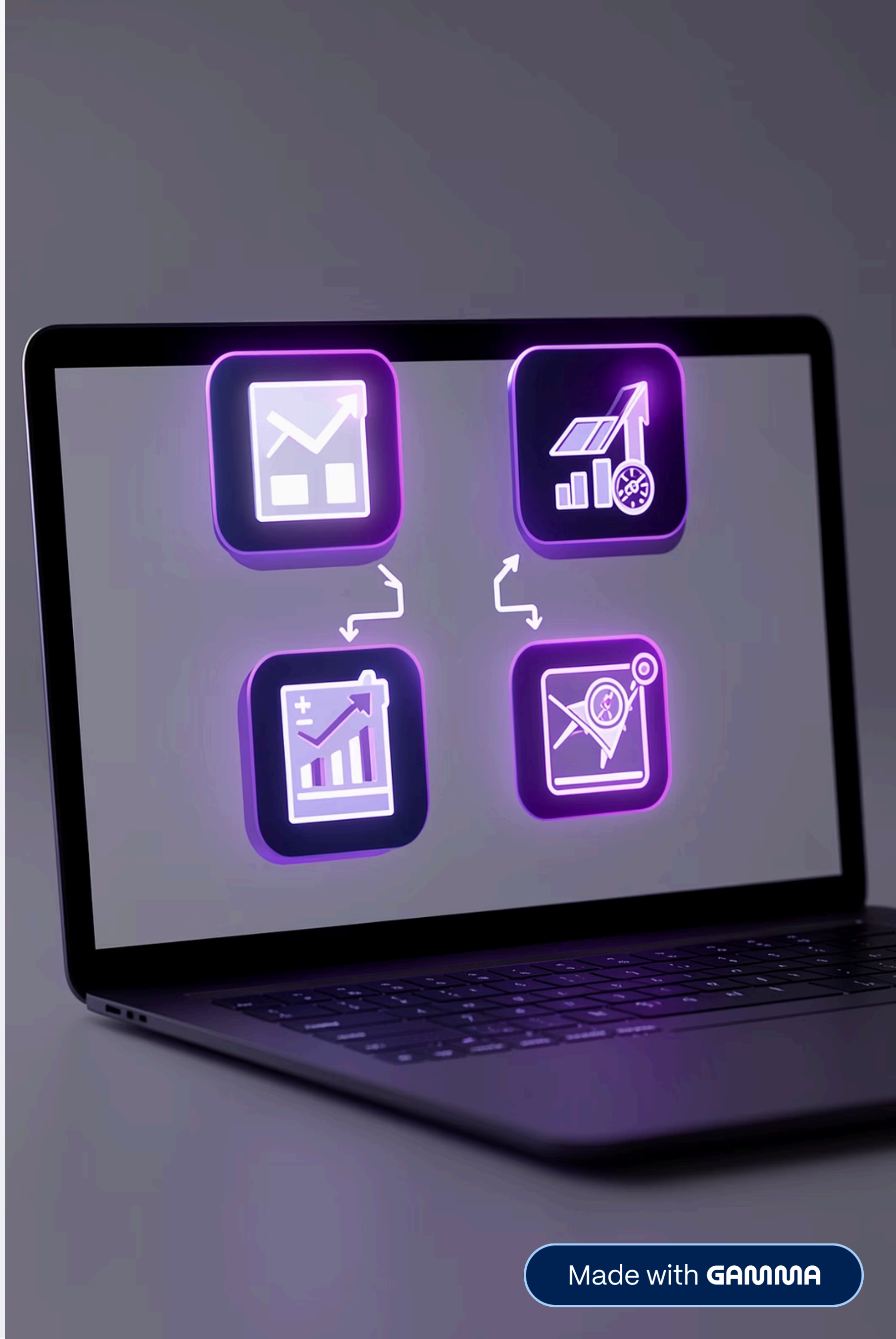
Calculate current academic grade from scores

📚 Books Don't Match

Compare datasets and identify mismatches

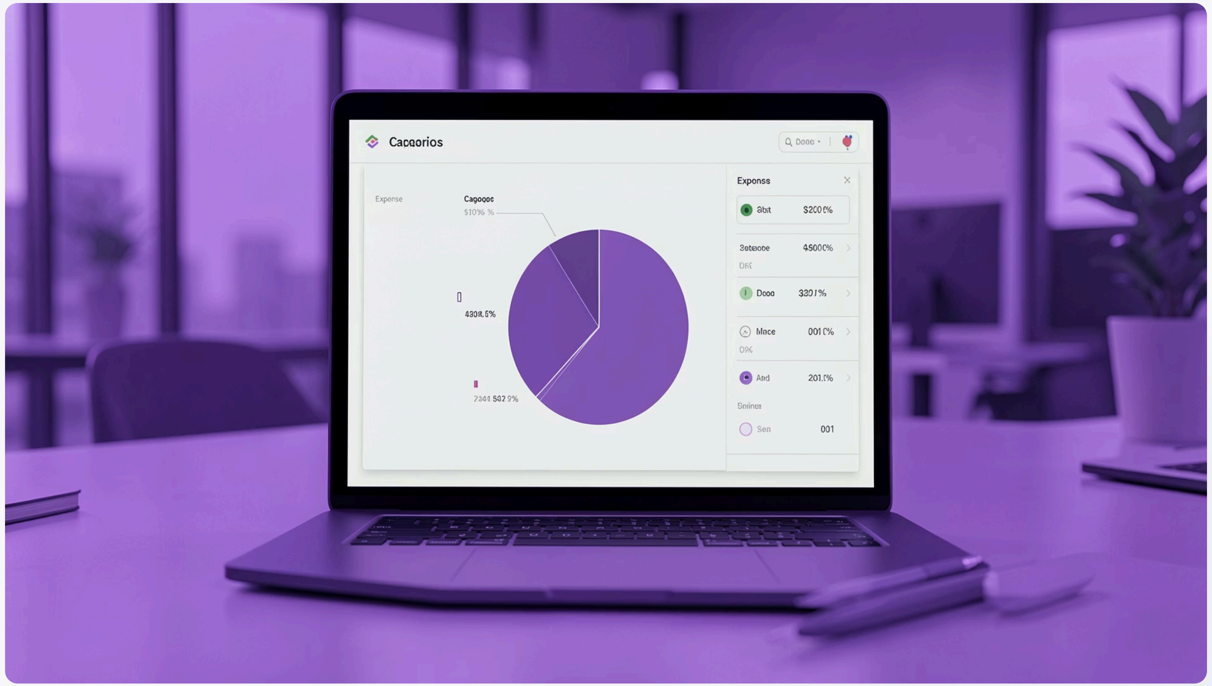
🗂️ Organize the Mess

Safely organize files without modifying originals



PROJECT 1

Money Detective



Problem Solved

Categorize monthly expenses, calculate totals, and identify the highest spending category.

AI Tool

ChatGPT

Initial Prompt

Write and run code to total expenses by category, find the biggest category, and show the exact total.

Improved Prompt

Explain the output and keep calculations accurate.

PROJECT 1 — RESULTS

Money Detective: Verification & Learnings

Verification

Compared AI-generated totals with manual calculations to confirm accuracy.

What Worked

AI generated correct Python code and clear output. Only minor prompt refinement was needed for better formatting.

Final Result

Successfully calculated all category totals and identified the highest expense category.

Key Takeaway

AI speeds up data analysis — but manual verification ensures accuracy.

PROJECT 2

What's My Grade?

1

Problem

Calculate current academic grade using specific grading rules

2

Prompt

Write a Python script that calculates current grade from scores, then refine for clean, explainable output

3

Verify

Manually verified result using the grading formula

4

Result

Successfully calculated the current grade after clearly providing the grading rules

Key Learning: Prompt engineering and validation of AI-generated calculations are essential skills.

PROJECT 3

Books Don't Match

Problem Solved

Compare two datasets and identify missing or mismatched records.

AI Tool

ChatGPT

Prompts Used

Initial: Compare these records and identify differences using Python.

Improved: Generate a clear comparison report and explain the mismatches.

Verification

Reviewed comparison output against the provided data.

PROJECT 3 — RESULTS

Books Don't Match: Outcome

What Worked

Comparison logic worked well. Minor prompt improvements made the report easier to read.

Final Result

Successfully detected all mismatched and missing records between the two datasets.

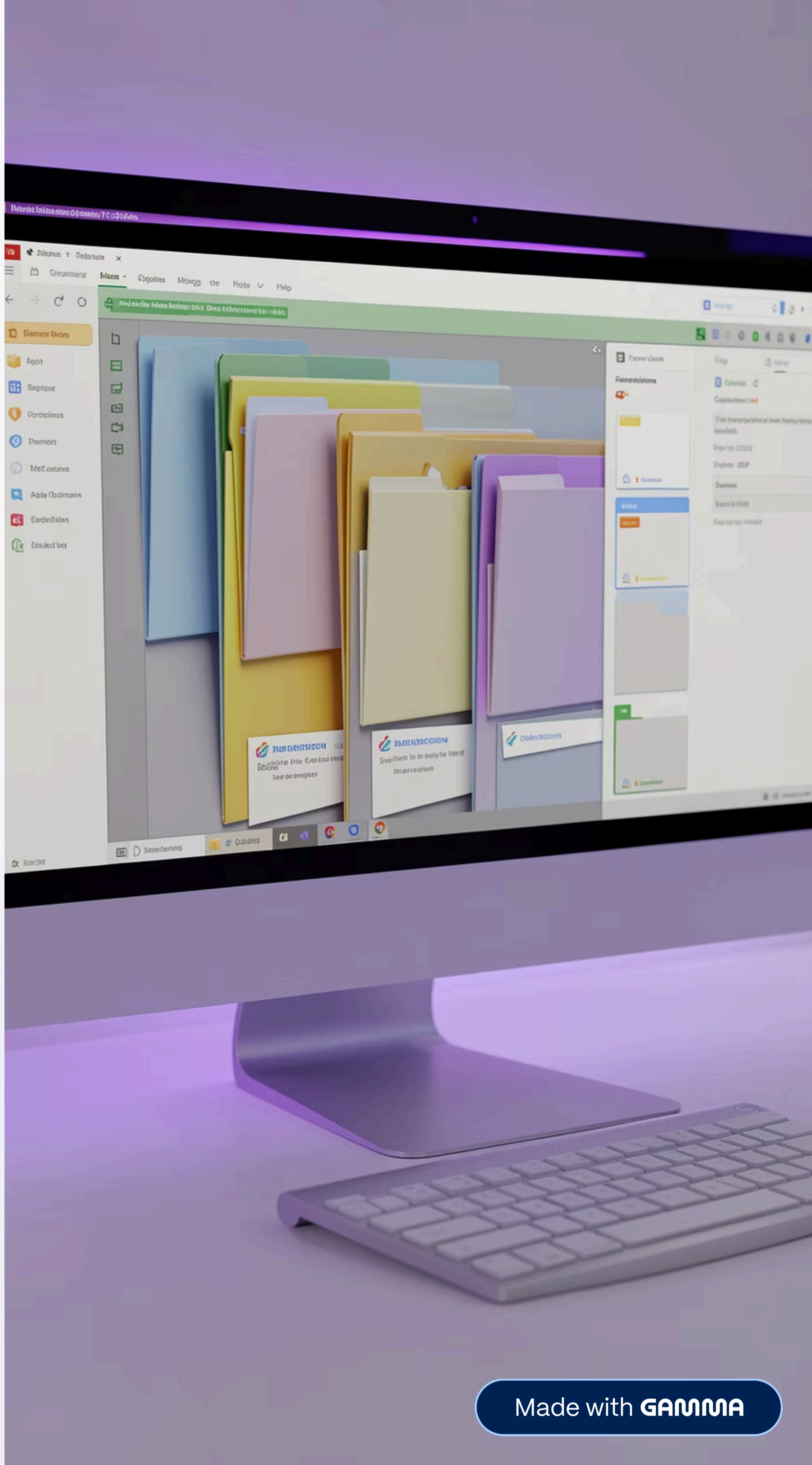
Key Learning

AI effectively assists in data comparison and reconciliation tasks when guided with clear prompts.

PROJECT 4

Organize the Mess

1	**Problem** Safely organize a copied folder without modifying the originals
2	**Initial Prompt** Organize files by type, detect duplicates, identify files larger than 100 MB, do not delete anything, and show a dry run first
3	**Improved Prompt** Show every file to be moved or renamed and wait for approval before execution
4	**Verification** Compared organized folder with original copy and confirmed nothing was overwritten



PROJECT 4 — RESULTS

Organize the Mess: Safe Automation

What Worked

Dry-run planning improved safety. No files were deleted.

Final Result

Files were organized into folders by type while fully preserving the original folder.

Key Learning

The importance of **backups**, **dry runs**, and **safe automation** when working with file systems.

Key Takeaways Across All Projects

Prompt Engineering Matters

Clear, specific prompts consistently produced better, more accurate code across all four projects.

Always Verify Manually

Every AI-generated result was checked by hand — a critical step that ensured accuracy and built trust.

AI Accelerates, You Decide

AI speeds up data analysis, comparison, and automation — but human judgment guides safe, responsible use.